

Jack (Yuanman) Mu

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EDUCATION

University of Southern California, Viterbi School of Engineering | Los Angeles, CA

Jan 2026 -

Master of Science in Computer Science

- **Core Courses:** Foundations of Artificial Intelligence, Database Systems

Boston University, College of Arts and Sciences | Boston, MA

Sept 2021 – May 2025

Bachelor of Arts in Computer Science, Minor in Applied Statistics | GPA: 3.56/4.0

- **Honor:** Dean's List (Fall 2023, Spring 2023, Fall 2022, Spring 2022)
- **Core Courses:** Generative AI for Software Development, Software Engineering, Data Structures, Algorithms, MEAN Stack Application Development, Mobile App Development, Computing Systems, Database Systems, Theory of Computation, Analysis of Algorithms, Data Science in R, Object Oriented Programming

PROJECTS

Full-stack Android App Development | Team Leader

Dec 2024

- Designed a responsibly designed full-stack Android App for English vocabulary learning using latest Jetpack Compose
- Used Moshi and Retrofit with Coroutines to handle API integrations, and used Firebase for authentication
- Implemented Room with cloud synchronization for real-time updates of studying progress across device
- Built UI with Material 3, multi-screen flows, LiveData, and ViewModel
- App tested, including Unit test, UI test, and Error Logging

Impact Analysis: Mayor's Office of Returning Citizens, City of Boston | Team Leader

May 2024

- Developed a comprehensive web dashboard analyzing the impact of the Office of Returning Citizens (ORC) in FY22-24
- Led weekly meetings, client meetings, data analysis, dashboard backend development, database schema
- Converted textual data into interactive numeric analysis using Python and NLP, producing charts and graphs for analysis
- Analyzed the statistical impact from the performance of ORC, and generated interactive graphs for ORC's annual report to the Mayor's Office

Racial and Economic Disparities Analysis in Food Accessibility | Team Leader

Dec 2023

- Analyzed food accessibility disparities across 72,532 U.S. census tracts using USDA's Food Access Research Atlas, reducing 147 variables by 60% through data cleaning
- Explored the relationship between racial demographics, poverty rates, and supermarket accessibility, identifying a notable 0.53 correlation between African American populations and poverty
- Integrated state GDP data to contextualize findings, improving model precision (R^2 increased from 0.19 to 0.24)
- Visualized key patterns via scatter plots, heatmaps, and regression analysis, highlighting systemic inequalities in food access and informing potential policy interventions
- Critically evaluated model limitations and proposed future work incorporating longitudinal and qualitative data

Spotify Library-Based Web Application | Team Leader

Dec 2023

- Collaborated with a team of 4 and developed a fully functional web application with a dynamic JavaScript-based front-end, an Express.js based back-end, a MongoDB database that stores back-end application state
- Coded most of the backend & API integration & database interactions

PROFESSIONAL EXPERIENCE

Client Services & Support | Boston University IT Help Center IS&T | Boston, MA

Sept 2023 - May 2025

- Provided technical support to students and staff, handled the delivery of devices around the campus, helped with troubleshooting hardware and software issues during deliveries
- Developed problem-solving skills and improved communication of technical information

Academic Mentor | Boston University Bridge to BU Program | Boston, MA

Sept 2021 - Dec 2022

- Supported international students' adjustment to BU through English practice, cultural discussions, and team activities
- Enhanced communication and leadership skills by guiding mentees in academic development

RESEARCH EXPERIENCE

Analysis of Machine Learning Techs | Nanjing, China

July 2023

- Conducted cooperative research analyzing the performances of CNN, ANN, GRU, LSTM, ARIMA, and Transformer models in forecasting stock prices
- Contributed to findings that indicated LSTM outperformed other algorithms
- Produced a paper "Comparative Analysis of Different Machine Learning Techniques in Forecasting Stock Price"

TECHNICAL SKILLS

Programming Languages: Python, Java, HTML, SQL (relational databases MySQL), schema design, JavaScript, R, Kotlin, C

Familiar with: Next.js, React, Node.js, Jetpack Compose, noSQL, MongoDB, DBMS, XML, AWS, Git, REST